

Expansion through Co-optation:

A Strategic Model of Conquest with First-Strike Advantage and Local Talent

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Abstract

The Mongols created the largest empire in human history with an undefeatable army. For over 80 years, the Mongols were in an everlasting campaign for territorial conquest, making themselves the most notorious for “wanton and senseless” (Kalyvas, 1999) “lamented barbarians” (Spruyt, 1994) in the historical record. Mongols initiated a large number of senseless, hard-fought battles in Eurasia. Was history made because bargaining tables were not invented for the uncivilized nomadic? Or were there strategic reasons for their violent record? This essay starts by trying to provide a rationalist answer to this phenomenon and develops a generic strategic model that captures the idealized dynamics under which the empire expands with coalitional politics.

Keywords: Balance of power, power-sharing, authoritarianism, co-optation, indirect rule.

1. Introduction

Two definitive criteria characterize nomadic conquest styles: their striking battlefield advantage and their lack of inbuilt bureaucratic apparatus. The incentive structure behind nomadic strategic decisions of conquest rests on these two premises. In my stylized model, we could see two stylized facts: first, relying solely on the possibility of exploiting local talent strongly predicts a tactful alliance or even a peaceful initial alliance. However, introducing a first strike advantage helps to accommodate the fact that the Mongols initiated so many ruthless campaigns against their target polities without any coalitional compromises.

With its unprecedented landscape and horror imbued in its “barbarian” legacy, the rise and fall of the Mongolian empire remain an enigma. The Mongols ruled an empire of an unparalleled scale for nearly 150 years, a feat far more enduring than a mere flash in the pan. This record is hardly a result of overkilling and blunders (Weatherford, 2005; Twitchett et al., 1978). And yet quite confusingly, a vast majority of the conquered regions were indeed catastrophically plundered and left misruled at the initial point of capture. Almost like a child keep buying expensive toys that they just destroy in the first use, Mongols seem to keep paying the grave cost of violent conquest over a flourishing polity without bothering to govern. This leaves many to conclude that they are simply maniacs who want to destroy. Potential strategic reasons supporting this record, moreover, are left under-theorized.

An alternative explanation starts by appreciating that Mongolians make decisions under circumstances quite different from those of the typical modern states sitting at the crisis bargaining tables

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today. Their chips are high on their striking battlefield advantage, but short on innate bureaucratic structure. Therefore, the cost-benefit analysis might tilt towards war. Being a people on horseback, Mongols are excellent at wielding special advantages in raiding tactics. In the famous battle of the Kalka River, for example, the Mongols feigned their escape and retreated to the steppe with their horses. For days, they slept on the horses and watched Rus' forces exhausted themselves trying to find the "defeated." As the Rus' were shattered by the week-long hunt, Mongols strikes again and demolished the coalition. Tactics like this creates crucial advantages in preemptive strikes of their time that cannot be cashed in a bargaining table. But upon central capture, Mongols do not have a team of administrative staffs to replace the conquered elites. Unlike later conquerors such as Britain, Mongols were in no way capable of installing an administrative apparatus locally and had to rely heavily on the conquered elites if they care to govern. In China, Mongols first left the northern part largely unrulled for half a century and then subsumed themselves under the remaining bureaucratic skeleton of Song's reign; In Iran, similar things happen when Mongols inherited the local apparatus from Persian traditions. The Golden Horde and Russia, on the other hand, were largely ruled indirectly through the tribute and tax farming system. Either way, Mongols relied heavily on local agents to establish a viable extractive apparatus for revenues.

Incorporating these two attributes, namely the outstanding battlefield advantage and dependence on local administrative skills, I developed a model of conquest with a coalitional twist to capture the Mongolian mode of expansion. When a first strike advantage is not present, the conqueror should have always collaborated with at least one of the domestic factions instead of fighting alone. However, introducing a first strike advantage in the formula supports a sensible reason why conquerors like the Mongols initiated so much unilateral plundering. The preemptive battlefield advantage for the Mongols creates a typical commitment problem for peaceful bargaining, and the fact that you cannot cash out this advantage at the bargaining table creates incentives for the conqueror to forfeit the opportunity of exploiting local knowledge in instilling extractive institutions.

2. Literature Review

Empire expansion can be pursued in a variety of ways. A forward-looking empire must conduct this expansionist agenda by considering who can and will serve as an ideal delegation for future extraction. Therefore, much scholarly effort has been devoted to explaining how monitoring difficulties and lack of information shape the incentive structure of the empire in choosing potential delegates to deploy, which grounds the foundation for the repeated principal-agent paradigm. For example, Gailmard (2024) uses the information problem to explain how British institutional arrangements, such as check-and-balance institutions, were set up in the United States. Against the backdrop of the typical principal-agent context, factors such as military intervention (Padró i Miquel and Yared, 2012) or climate shocks (Garfias and Sellars, 2021, 2022) also help to explain variations of peace and conflict in colonial contexts.

These crucial trade-offs in the elite delegation problems and different institutional arrangement choices might also occur at an earlier stage, when emperors decide whether to co-opt a domestic faction to aid them on the battlefield and what subsequent administrative apparatus can be upheld in their conquering period. More specifically, the emperor must decide several things in initiating conquest: (i) whether to take over a polity; (ii) which agent to delegate administrative power; and (iii) how to best elicit the best effort from the chosen agent. The principal-agent literature provides ample answers for the last part of the equation, but few studies go back to decisions in the (i) and (ii) stages. Typical crisis bargaining literature, moreover, does not break into the sub-national

context and engage with different institutional choices.

The key insight of coalitional politics is pioneered by the winning coalition dynamics (Bueno de Mesquita and Lalman, 1988, 2008) decision to co-opt factions in multiple party conflicts parallels the dynamics of coalitional politics and balance of power literature in classic works of international relations. A bilateral coalition is always unstable once the third party is ousted from the horizon. This expectation would then induce the stability of a three-player coalition, as internal fragmentation would create subsequent survival threats. A significant development and application of this vein is found in Acemoglu et al. (2012), where future stability concerns undermine the possibility of pursuing more efficiency-inducing arrangements today. The strategic actions include forming coalitions, altering “constitutions”, or excluding members, all with an eye to future advantages. A more general analysis of multipolarity is offered by Wagner (1986), where systematic stability or conflict is mechanically linked with the number of players available and the balance of power within. The model’s central tactic is that states form or abandon coalitions based on expected power distribution shifts and the anticipated outcome of conflict or negotiation. The causal mechanism driving behavior is the forward-looking expectation of whether one’s position will improve or deteriorate over time: a declining power has incentives to initiate conflict or seek allies to hedge against loss, while a rising power may prefer to delay, negotiate, or bandwagon later. Out of these key theoretical works, the three-player setup is often the minimum level of complication to capture the central dynamic, which shall serve as the basis for my model.

As insightful as the systematic analyses are, their results are not sensitive to the specific power dynamics of the players, or are only sensitive to some discrete version of it. Given different level of relative power dynamic, weaker and stronger players behave differently when interacting with each other. This key trade-off in coalition politics is highlighted by Morrow (1991). In contrast to the assumption that there is straightforward capability aggregation in alliance formation, Morrow (1991) highlights that a security autonomy trade-off exists within any coalitional pair. The strong have ground to demand concessions from the weak, which is counterintuitively inductive to the formation and sustaining of key coalitions. In a more symmetric coalition, the parametric threshold of interest alignment to sustain mutual commitment is higher, making a sustainable coalition of this sort less likely. Morrow (1993) further developed this rationale, arguing that arms building and alliance seeking all fall within the same budget constraint of domestic political cost. In particular, he argues that Austria and France failed to unite against the mutual threat of Prussia precisely because arming separately is more cost-efficient in domestic political terms.

Incorporating the continuous dimensions of power dynamics, others focus on the collective action dynamics in this analysis. The fact that investment in coalitions incurs both public good provision and private good provision creates tension under which the weaker or stronger faction might contribute disproportionately less or more (Olson and Zeckhauser, 1966). Because of the income effect, Olson and Zeckhauser (1966) finds that a larger faction of a coalition would bear a disproportionately greater share of the cost compared to its share of the benefit.

Later models by Niou and Zeigler (2019) and Niou and Ordeshook (1990) qualify the central logic of this collective action rationale, as they find that it is the smaller faction in a coalition that bears a disproportionately high cost in delivering coalitional dominance against a third party. In their model, two states face an external adversary but also distrust each other, characterized by a double competition within the coalition against a third party. They also explored the dynamics of balancing or bandwagoning. In situations where three players are at stake, they verified the long-standing conventional wisdom that peculiar distribution of power would override history of interaction, where states may ally with a longtime rival rather than a trusted friend if that rivalry,

by spurring commitment, yields a more effective coalition. In their latest analysis, Niou and Zeigler (2024) further discusses more ambiguous arrangements than committed alliances.

The timing of coalition formation is also a key point of contention. Wolford (2015) offers formal models examining how alliances operate and evolve during wartime, focusing on the formation, expansion, and breakdown of coalitions in conflict. The central mechanism is each state’s calculation of its war aims and the post-war balance: potential allies decide whether to enter the war based on how victory or defeat would alter their future power or spoils. Wolford finds that if a potential ally shares the goals of the defending side, dominating the status quo in fear of the aggressor’s victory, it will be balanced by joining the coalition to stop the aggressor, consistent with traditional balancing theory. However, suppose a third party sees an opportunity to profit from someone else’s war. In that case, it might bandwagon with the aggressor or demand concessions to stay out. This captures a coalition dynamic where not all powers automatically join to balance, echoing insights by Schweller (1994) that some states bandwagon for profit rather than balance for security.

Balance of power dynamics and coalition politics are integral parts of the classic international relationship theories (Waltz, 2001; Morgenthau, 1948; Walt, 1985), where a significant portion of the theoretical endeavor has been devoted to a three-player setup with an eye on bandwagoning or balancing results. However, this three-player setup is not limited to the interaction between national states. Weingast (1997)’s insight on sovereign transgression over private interest highlights the fruitfulness of applying these insights in a domestic setup. Echoing recent scholarly attempts to analyze elite politics within a domestic context during colonial times (Spencer Hartnett and Saleh, 2024), I develop a model of colonial expansion and co-optation.

3. The General Context

One of the most fundamental insights developed to explain nomadic conquest is led by the seminal work by Barfield (1989a) and later work of Turchin (2009). The domestic conditions of a typical nomadic life and class struggle alone typically fail to induce military concentration. As Barfield (1989b) points out, the making of the Mongolian empire deviates from typical successions and tribal quarrels in the steppe. What is the source of this deviation? The “perilous frontier” at the Eurasian steppe creates a lucrative chance for bandits for the Mongols to rob their wealthy neighbor, China. Through repeated raids, negotiations, trades, and tribute arrangements, the Xiongnu could feed a large population that would not have survived if confined to the steppe. Over the years, however, an increasingly centralized and more assertive China could afford better defenses against nomadic hassles. The nomadic people, the story goes, are left hungry for the abundant resources typically offered by their agrarian neighbor. Similar to all the other prominent empires in the Eurasian steppe, the Mongols exploited their horse-raid and military dominance expertise to assert favorable trade treaties or outright extortion out of the available agrarian societies.

However, an unprecedentedly successful military centralization made the Mongols capable of handling more than a tribute deal. Unlike previous dealings between nomads and China, the Mongolian empire did not stop at loose coalitions within the tribes. Instead, they were able to wield a highly centralized people with uniform mobility and martial readiness. As a result, they were not content with setting up ad hoc tribute treaties with important polities by staying at the periphery (Weatherford, 2005; Barfield, 1989a). Instead, they started immigrating from the Eurasian steppe core to the newly conquered polities and establishing rule there.

Consideration of subsequent ruling arrangements becomes more pressing at this stage. Whether

conquest can be manufactured highly depends on the availability of a trustworthy agent in the local context. Without defection from administrative teams in China, the Mongols lacked their own administrative structures to govern an agrarian empire properly. The excessive mistrust of Genghis Khan in any form of delegation of power, even to his families (Weatherford, 2005; Barfield, 1989a), however, forbids the unrestricted use of agents of this sort. In the meantime, Kublai Khan patiently developed tactics other than brute force in the Mongolian toolbox of conquest. He inherited attempts by Ogedei Khan to instill a stationary capital in current Beijing and worked to develop an extractive apparatus suitable for the agricultural context.

4. Model Setup

There are three actors, a foreign force (F), and two domestic factions, D_1 and D_2 , within a target state. D_1 's share of the status quo within the target is $s \in (0, 1)$ and D_2 's share of the status quo within the target is $1 - s$. F contemplates a possible takeover of the target territory via unilateral invasion or co-option that can result in indirect or direct rule after the takeover. Domestic factions can cooperate with the external power or resist foreign invasion. The sequence of play, visualized in Figure 1, is as follows.

1. F chooses to maintain the status quo, invade the target territory unilaterally, or co-opt the domestic factions by proposing $\tau = (\tau_1, \tau_2)$, $\tau_i \in (0, 1)$. Once the proposals are made, they are common knowledge for both factions.
2. If F chooses co-option, D_1 and D_2 simultaneously choose to cooperate with or resist the foreign power.
3. Alliance and fighting take place, and payoffs are realized.

Each actor is endowed with military capability $m_i > 0$, for $i \in \{f, 1, 2\}$. In general, the probability of winning by a side with military capability x against another side with military capability y is $p(x, y) = \frac{x}{x+y}$, which increases in x and decreases in y . Observe that $p(x, y)$ is concave in x and convex in y .

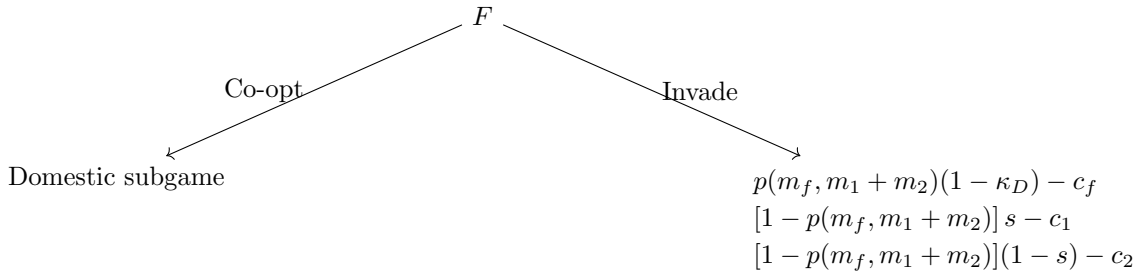


Figure 1: Game tree.

The specific payoff structure within this stream of game play is set as follows. If F chooses to maintain the status quo, it receives 0; D_1 and D_2 receive their corresponding share of the status quo $s_1 = s$ and $s_2 = 1 - s$, respectively. If F invades unilaterally, it faces the collective resistance of D_1 and D_2 , and implements direct rule if it wins the war. War is costly, and we assume that each

	D_2 Cooperate	D_2 Resist
D_1 Cooperate	Indirect rule, two vassals	F & D_1 vs. D_2 ; indirect rule via D_1
D_1 Resist	F & D_2 vs. D_1 ; indirect rule via D_2	Collective resistance; direct rule if F wins

Table 1: Domestic subgame.

actor also pays a cost c_i , $i \in \{f, 1, 2\}$ for fighting. Therefore, F 's expected utility from unilateral invasion is

$$U_f(I) = p(m_f, m_1 + m_2)(1 - \kappa_D) - c_f, \quad (1)$$

where $\kappa_D > 0$ represents the costliness of the direct rule. Technically, this term can be absorbed into the pie or probability of war, but it would be more intuitive to single out this term for substantive discussion in later comparative statics sections. A higher κ_D means a higher cost of direct rule, which might involve administrative costs and the costs of repressing mass resistance against foreign rule.

If the foreign invasion fails, the domestic factions maintain their rule according to the same distribution of benefits prior to the fighting. Thus, D_1 's expected utility if F chooses to invade is

$$U_1(I) = [1 - p(m_f, m_1 + m_2)]s - c_1,$$

and similarly, D_2 's expected utility if F chooses to invade is

$$U_2(I) = [1 - p(m_f, m_1 + m_2)](1 - s) - c_2.$$

If F chooses to co-opt, D_1 and D_2 simultaneously choose to cooperate with or resist the foreign power. If both factions cooperate with the foreign power, F implements an indirect rule with two vassals with an expected utility of $\tau_1 s + \tau_2(1 - s)$. The payoffs for D_1 and D_2 under indirect rule are $(1 - \tau_1)s$ and $(1 - \tau_2)(1 - s)$, respectively. If both factions choose to resist F , then collective resistance takes place and the expected utilities are similar to those when F chooses unilateral invasion. The only difference is that now F does not enjoy the first-strike advantage as it loses the chance of a surprise attack. Thus, if F proposes any (τ_1, τ_2) that is rejected by both domestic factions, F receives

$$p(m_f, m_1 + m_2)(1 - \kappa_D) - c_f,$$

and D_i receives $[1 - p(m_f, m_1 + m_2)]s_i - c_i$.

If one faction cooperates while the other faction resists, the foreign power allies with the cooperative faction to fight against the resisting faction and implement indirect rule through the cooperative faction upon victory. Thus, F 's expected utility from D_i cooperating and D_j resisting is

$$p(m_f + m_i, m_j)\tau_i - c_f.$$

The cooperative faction D_i 's expected utility is

$$p(m_f + m_i, m_j)(1 - \tau_i) - c_i.$$

If the resisting party wins, it will rule alone. Therefore, the resisting faction D_j 's expected utility is

$$1 - p(m_f + m_i, m_j) - c_j.$$

5. Equilibrium Analysis

Proposition 1 (Optimal Co-optation Proposals). *Let $\underline{s} \equiv p(m_1, m_2) - \frac{c_f + c_1 + c_2}{p(m_1 + m_2, m_f)}$ and $\bar{s} \equiv p(m_1, m_2) + \frac{c_f + c_1 + c_2}{p(m_1 + m_2, m_f)}$. The following holds if F chooses to co-opt the domestic factions.*

1. *If $s < \underline{s}$, F offers $\tau_1 = \tau_1^R$ and any $\tau_2 > \max\{\tau_2^R, \tau_2^C\}$. D_1 cooperates and D_2 resists.*
2. *If $s > \bar{s}$, F offers $\tau_2 = \tau_2^R$ and any $\tau_1 > \max\{\tau_1^R, \tau_1^C\}$. D_2 cooperates and D_1 resists.*
3. *If $s \in (\underline{s}, \bar{s})$, F offers (τ_1^C, τ_2^C) , and both D_1 and D_2 cooperate.*
4. *Mutual resistance is not in equilibrium.*

Proposition 1 describes F 's optimal choice of co-optation proposals if it decides to implement indirect rule via one or both of the domestic factions. Figure 2 visually presents optimal indirect-rule proposals under different distributions of the domestic status quo, s . If $s < \underline{s}$, that is, D_1 's share of the status quo is much smaller than its share of military capability against D_2 , then F will choose the maximum level of indirect rule that just induces D_1 to cooperate and propose any degree of indirect rule that induces D_2 to resist. Similarly, if $s > \bar{s}$, that is, D_1 's share of the status quo is much larger than its share of military capability against D_2 , then F will choose the maximum level of indirect rule that just induces D_2 to cooperate and propose any degree of indirect rule that induces D_1 to resist. If $s \in (\underline{s}, \bar{s})$, that is, both factions' share of the status quo is more or less balanced relative to their military capability, then F propose the maximum level of τ that induce both factions to cooperate and implement indirect rule through two vassals.

Highlighted by Acemoglu et al. (2012), one might overestimate how stable a two-party coalition is in anticipation of within-coalition conflict after the third party is out of the picture. Considering the aftermath, Foreign can only promise the collaborating faction that the only viable offer is its war payoff.

Lemma 1. *The offer given to the single faction is always enforceable, considering expected intra-coalition instability.*

Lemma 2. *If $1 - \frac{c_1 + c_2 + c_f}{p(m_1 + m_2, m_f)} < p(m_1, m_2) < \frac{c_1 + c_2 + c_f}{p(m_1 + m_2, m_f)}$, Foreign always offer both factions a co-optation deal.*

This lemma speaks to some of the key findings in systematic analysis about coalitions. When $p(m_1 + m_2, m_f)$ decreases, or when the alliance's aggregate capability decreases relative to Foreign, the left side will be lower, and the right side will be higher. This entails a broader range of peaceful bargains for the potential distribution of military endowments that could fall within it. The key parameter is $p(m_1, m_2)$, which falls within a moderate range if and only if the two domestic factions are of relatively equal power. When the relative dominance of Foreign over the aggregated capacity of the two factions is sufficiently overwhelming such that $p(m_1 + m_2, m_f)$ is sufficiently small, any distribution of military endowment within the polity would induce peace, confirming the key insight offered by Wagner (1986); when that dominance is moderate, a sufficiently balanced share of the military endowment would also pacify potential defections. Notice that this peace condition is not sensitive to the share of domestic spoils (s), but is a rather mechanical result of the "hard" balance of power distribution.

In sum, if it is an uneven distribution of rents relative to their military power among domestic factions, the foreign power will always pick the domestic faction that enjoys a disproportionately small share of the status quo relative to its military capacity, the unsatisfactory faction, as the vassal, ally with it to fight the other faction, and implement indirect rule via that faction upon victory. By contrast, the foreign power will implement indirect rule via multiple vassals if none of the domestic factions is particularly grieved. Because war is destructive, the foreign power has to weigh the cost of war against the benefit of additional rents. If the cost of fighting is sufficiently high, the range of moderate s becomes larger as the cost of war for any party increases, and Foreign has incentives to co-opt both factions to avoid the costly war.

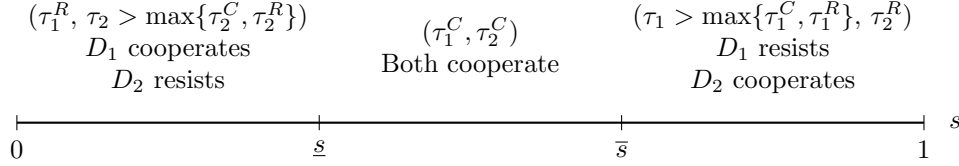


Figure 2: Foreign's optimal co-optation proposal across the share-of-status-quo s .

Proposition 2 (Foreign Co-optation vs. Unilateral Invasion). *Foreign never chooses unilateral invasion.*

Proposition 2 highlights a significant result. Foreign should always choose a tactful collaboration with at least one of the domestic factions for better chances of winning the war or exploiting the peace premium. Initiating unilateral invasion by and of itself is always dominated, for it incurs an additional cost of direct rule (or, in a technical sense, a higher cost of war). But we know that historically, Mongols often resorted to raiding and plundering over a polity, do we have to conclude that their conquest decisions were not as rational and calculated?

6. Extension: Battlefield Surprise Advantage

In this section, I introduce Mongolians' particular battlefield advantage to reconcile the fact that there were so many unilateral invasions without co-optation in the Mongolian history of conquest.

Assume that if F chooses unilateral invasion, it enjoys a first-strike advantage that captures the element of surprise in preemptive attacks, where F can cash out some battlefield advances from the domestic factions' unpreparedness. I capture this by introducing a parameter λ , so that when F chooses to take unilateral military actions against the domestic factions, it wins with probability $p(\lambda m_f, m_1 + m_2)$, where $\lambda > 1$ reflects the degree of first-strike advantage by F . A larger λ implies a higher degree of first-strike advantage by F . Straightforwardly, we could see that the optimal play at the domestic subgame remains the same, and the only part getting changed is the expected payoff for the unilateral invasion scenario. More specifically, now the incentive structure of all the parties involved is adjusted as follows.

$$U_f(I) = p(\lambda m_f, m_1 + m_2)(1 - \kappa_D) - c_f, \quad (2)$$

$$U_1(I) = [1 - p(\lambda m_f, m_1 + m_2)]s - c_1, \quad U_2(I) = [1 - p(\lambda m_f, m_1 + m_2)](1 - s) - c_2.$$

Proposition 3 (Foreign Co-optation vs. Unilateral Invasion under First-Strike Advantage). *In equilibrium, we have the following conditions.*

1. If $s < \underline{s}$, there exist $\kappa_D^\dagger > 0$ and $\lambda^\dagger > 1$ such that F chooses unilateral invasion if and only if $\kappa_D < \kappa_D^\dagger$ and $\lambda > \lambda^\dagger$. Otherwise, F proposes $\tau_1 = \tau_1^R$ and any $\tau_2 > \max\{\tau_2^R, \tau_2^C\}$; D_1 cooperates and D_2 resists; F implements indirect rule via D_1 upon victory.
2. If $s > \bar{s}$, there exist $\kappa_D^\ddagger > 0$ and $\lambda^\ddagger > 1$ such that F chooses unilateral invasion if and only if $\kappa_D < \kappa_D^\ddagger$ and $\lambda > \lambda^\ddagger$. Otherwise, F proposes any $\tau_1 > \max\{\tau_1^R, \tau_1^C\}$ and $\tau_2 = \tau_2^R$; D_1 resists and D_2 cooperates; F implements indirect rule via D_2 upon victory.
3. For $s \in (\underline{s}, \bar{s})$, there exists $\tilde{\kappa}_D > 0$ and $\tilde{\lambda} > 1$ such that F chooses unilateral invasion if and only if $\kappa_D < \tilde{\kappa}_D$ and $\lambda > \tilde{\lambda}$. Otherwise, F offers (τ_1^C, τ_2^C) , both D_1 and D_2 cooperate, and F implements indirect rule via two vassals.

This proposition reconciles the fact that Mongols initiated so many unilateral conquests instead of appealing to a British balance-of-power style of conquest. Even when the cost of direct reign is high to the extent that the Mongols are supposed to appreciate a chance to utilize local talent, their stunning battlefield advantage may point them to brute force.

7. Empirical Context: 124 Mongol Conquests

This section pairs the model’s comparative statics with a hand-coded dataset of 124 Mongol conquest events between 1211 and 1302, cross-referenced from Jaques (2006) and Chandler and Fox (1974). Each event records the city, year of conquest, latitude and longitude, population at the time of capture, and the mode of subsequent Mongol rule, classified into three categories: *plunder* (the city is sacked but no administrative apparatus is installed), *indirect rule* (Mongols install an extractive arrangement that delegates day-to-day governance to local elites or tribute-paying intermediaries), and *direct rule* (Mongols install their own administrative cadre, typically by absorbing a pre-existing bureaucracy). The classification follows Weatherford (2005), Barfield (1989a), and the secondary historiography on each region.

7.1. Comparative Statics of the Model

Figure 3A plots Foreign’s optimal regime in the (κ_D, λ) space, holding $m_f = m_1 = m_2 = 1$, $c_i = 0$, and $s = 0.5$ fixed. The black curve marks the threshold $\lambda^* = \frac{2}{2-3\kappa_D}$ derived from setting $U_f(I) = U_f(\text{co-optation, dual})$ in this symmetric case. Above the curve and to the left, Foreign chooses unilateral invasion. Below the curve and to the right, Foreign chooses co-optation. Two patterns emerge. First, even modest first-strike advantage (λ slightly above 1) is sufficient to push Foreign to invade when the cost of direct rule is low. Second, as κ_D approaches $2/3$, the threshold λ^* diverges to infinity: no first-strike advantage can rationalize unilateral invasion when direct rule is sufficiently costly. The parameter region in which Mongol-style unilateral plunder is rational is therefore the southwestern corner of the figure: cheap direct rule, high first-strike advantage.

7.2. Distribution of Rule Modes

Figure 3B reports the count of cities by mode. Of the 123 cities with a definite classification, 65 (53%) were plundered without subsequent administrative installation, 28 (23%) were placed under indirect rule, and 30 (24%) were placed under direct rule. The plurality of conquest events did not

produce an administrative arrangement at all, which is the empirical fact the model is designed to rationalize.

7.3. Temporal and Geographic Patterns

Two empirical regularities sharpen the model’s interpretation. First, the mode of rule shifts away from plunder over time. Table 2 splits the dataset at 1240, the conventional break between the Mongols’ first wave of expansion under Genghis Khan and the second wave under his successors. Plunder accounts for 63% of conquests in the early period and 40% in the late period; indirect rule rises from 16% to 31%; direct rule rises modestly from 21% to 29%. The shift is consistent with the comparative statics in Proposition 3: as the Mongols’ first-strike advantage λ depreciated through prior exposure of target polities and as the supply of co-optable administrative pipelines (Khitan, Persian, Han) grew, the relative payoff to plunder fell.

Second, mode of rule tracks the size of the conquered city. Direct-rule cities have a median population of 50,000, indirect-rule cities have a median of 20,000, and plundered cities have a median of 10,000. The five-fold gap between direct-rule and plundered cities is consistent with the model’s mechanism: direct rule pays only when the conquered polity supplies a functioning administrative apparatus that the conqueror can subsume, and large cities are precisely the polities most likely to have one. The Mongols installed direct rule on Hangzhou and Tabriz, where the inherited Song and Persian bureaucracies were doing most of the day-to-day work; they plundered Bukhara and Urgench, where there was less to inherit and a higher first-strike payoff to extract on the way through.

Figure 4A shows the year-by-mode histogram with a vertical break at 1240. Plunder concentrates densely in the early decades, particularly during the 1219–1221 Khwarezm campaign and the 1236–1242 western expansion. Indirect and direct rule rise steadily after 1240 and dominate the later record. Figure 4B maps the geographic distribution of the three modes. Plunder events cluster in the central Eurasian corridor stretching from the Caucasus through Khwarezm to north China; direct rule concentrates in north China, the Persian plateau, and the Crimean and Volga zones where pre-existing administrative apparatuses were available; indirect rule is geographically intermediate and appears most often in the steppe-frontier zones where Mongol garrison authority overlapped with semi-autonomous local elites.

The two regularities together support the model’s interpretation. The early-period plunder concentration corresponds to the parameter region in Figure 3A with high λ and low κ_D ; the post-1240 shift to indirect and direct rule corresponds to the slow movement of the Mongol position into the co-optation region as the first-strike depreciation and the discovery of inheritable administrative apparatus reduced the relative payoff to unilateral invasion. The case study of Georgia in the next section examines a single trajectory of this shift in detail.

8. Substantive Application: Mongol Conquest of Georgia and Beyond

The Mongol campaigns in Georgia between 1220 and 1243 and subsequent rule provide an empirical case for analyzing the strategic dynamics formalized in the model. In particular, this case helps illustrate how a conqueror (F) with substantial first-strike advantage and lacking internal administrative capacity might oscillate between unilateral invasion and indirect rule via co-optation of local elites.

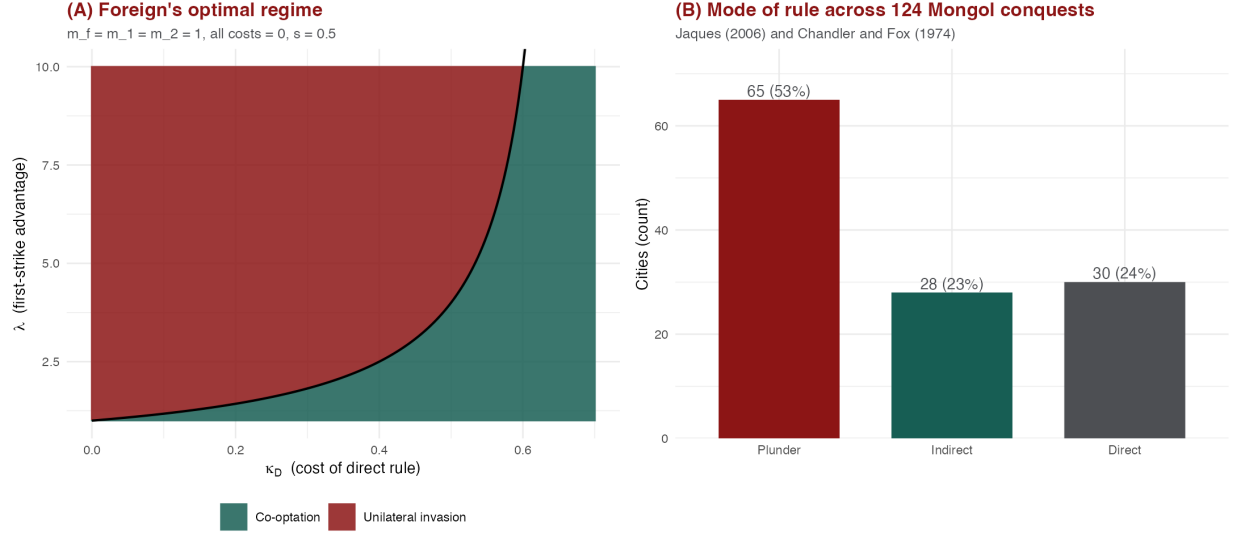


Figure 3: Panel A: Foreign's optimal regime in the (κ_D, λ) space, fixing $m_f = m_1 = m_2 = 1$, $c_i = 0$, $s = 0.5$. The black curve is the threshold $\lambda^* = 2/(2 - 3\kappa_D)$. Panel B: Distribution of rule modes across 124 Mongol conquests, 1211–1302. Source: hand-coded from Jaques (2006) and Chandler and Fox (1974).

Table 2: Mode of rule by period.

Mode	Early (1211–1239)		Late (1240–1302)	
	Count	Share	Count	Share
Plunder	43	63%	22	40%
Indirect rule	11	16%	17	31%
Direct rule	14	21%	16	29%
Total	68	100%	55	100%

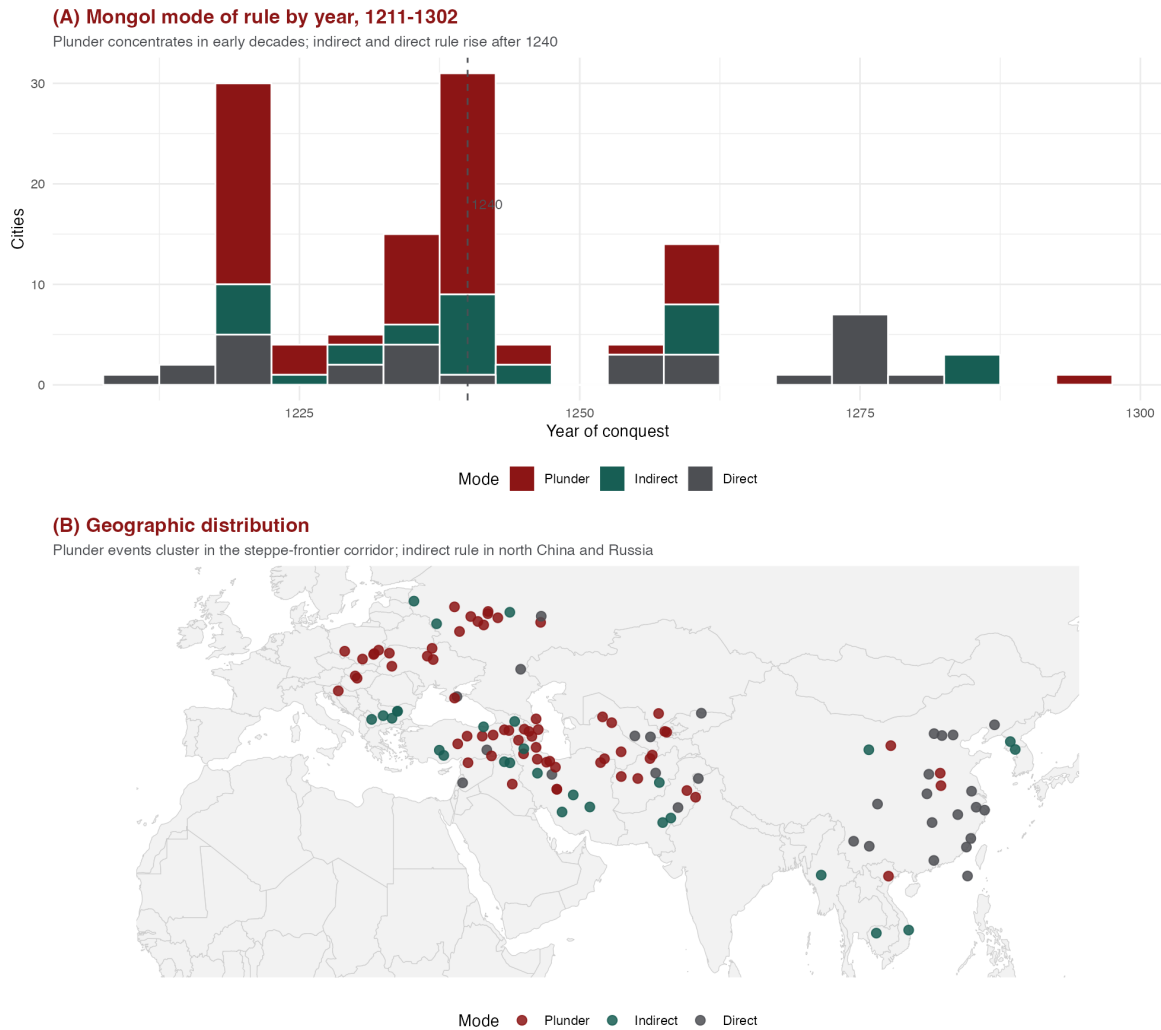


Figure 4: Panel A: Mongol mode of rule by year of conquest, 1211–1302. The dashed vertical line marks 1240. Panel B: Geographic distribution of the three modes. Source: hand-coded from Jaques (2006) and Chandler and Fox (1974).

8.1. Phase I: Surprise Plunders

The initial Mongol raids into Georgia and surrounding regions in the early 1220s resemble the model's unilateral invasion path under high first-strike advantage ($\lambda \gg 1$). The Mongols' initial contact with Georgia was mysterious. They initiated a series of surprise attacks in the peripheral regions of Georgia, including Caspian Sea, Kotman River, northeastern Armenia, Shirvan, and Kalka River. The regions were underprepared and were subject to easy plunders. Importantly, this series of surprise attacks left the core leaders of the Georgians in confusion as to the identity of their attackers. In particular, the record of one contemporary chronicler indicates that he is unaware of the nature of the attackers and does not mention them by name. In 1223, when the Mongols had seemingly deferred their plans regarding Georgia, King George IV's sister and successor Queen Rusudan wrote in a letter to Pope Honorius III, that the Georgians had presumed the Mongols were Christians because they fought Muslims, but they had turned out to be pagans.

8.2. Phase II: Fragmentation and Co-optation

By 1236, the battlefield records created by this series of surprise attacks have compelled the local elites to lean towards defection. Many nobles defected or passively submitted, while others resisted from fortresses. This empirical record echos the comparative statics of widespread peaceful co-optation. As the military dominance of the Foreign becomes apparent, the parallel co-optation scenario is more likely. Little by little, it became obvious that defection to Mongols is the dominant strategy for everyone, they received very little resistance everywhere. Upon seeing the defection of most princedoms since 1236, Atabek Avag negotiated and arranged Georgia's official submission to the Great Khan in 1243.

This divide and conquer dynamic is carried well into how the Mongols established subsequent indirect rule arrangements. Exploiting the complicated issue of succession, the Mongols had the Georgian nobles divided into two rival parties, each of which advocated their own candidate to the crown. These were David VII "Ulu", an illegitimate son of George IV, and his cousin David VI "Narin", son of Rusudan. After a failed plot against the Mongol rule in Georgia (1245), Güyük Khan made, in 1247, both pretenders co-kings, in eastern and western parts of the kingdom respectively. The system of *dumans* was abolished, but the Mongols closely watched the Georgian administration in order to secure a steady flow of taxes and tributes from the subject peoples, who were also pressed into the Mongol armies.

8.3. General Mongolian Conquest Patterns

This divide and conquer logic is also present in China. Mongolian ambition to expand the empire to the larger part of China converges with the Song's consistent but constant defeat in regaining lost territory from the Jin. Mongolian emperors first allied with the weaker faction and then opted to succeed as well. In 1227, a small troop ordered by Genghis Khan went into the Song territory in an attempt to conquer the Jin. This incident, however, taught Temujin that some territories controlled by the Southern Song might be an integral part of a feasible roadmap to eliminate the Jin dynasty. After some defeats in escalated armed conflicts from 1230 to 1231, the Song caved in for cooperation with the Mongolian power in 1233. Song's general, Meng Gong, played a crucial part in the Mongolian termination of Jin's reign, which lasted more than a century.

However, there is no subsequent indirect rule through elites being established. China initiated sev-

eral short-lived, successful campaigns against the Mongols, for Kublai was distracted by succession struggles against Ariq Böke (Twitchett et al., 1978). This initial scale escalated into 45 years of conflict and a prolonged attempt by Kublai to take over the Southern Song.

In general, the Mongols often use a “tsunami” tactic for expansion that exploits fragmented territories piecemeal (May, 2016). Typically, troops would surprise the primary target with a lightning war while pacifying adjacent territories to formulate meaningful unifications. That is, they are exploiting the collective action problem within the domestic factions with their speedy attack tactics.

9. Conclusion

Real-world conquests decisions are messy, subject to idiosyncratic matches and mismatches of personal quibbles. This model highlights idealized trade-offs in who to co-opt as the ruling agent, who to ally with in the battle, and how to cash out battlefield advantage in an expansion enterprise.

This paper develops a strategic framework to rationalize the Mongol Empire’s seemingly erratic and violent conquest behavior through a lens of coalitional dynamics and battlefield advantage. By embedding a first-strike advantage into a model of co-optation, the analysis sheds light on why Mongols often opted for unilateral plunder rather than the more institutionally stable paths of coalition-building and indirect rule. The model demonstrates that when surprise military dominance is sufficiently large and the costs of direct rule are low, unilateral invasion becomes a rational strategy despite its long-term inefficiencies.

The theoretical contribution shows how variations in domestic power distribution and conquest costs jointly shape the conqueror’s choice between plunder, indirect rule via selective co-optation, and inclusive governance through multiple vassals. The equilibrium patterns reflect familiar logics from coalition theory and commitment problems, but they adapt them to the imperial context of weak bureaucratic capacity and high mobility warfare.

Empirically, the model finds support in the Mongol campaigns in Georgia and China, where conquest decisions reflected a calculated mix of shock warfare and elite co-optation depending on local conditions. Ultimately, this essay underscores that Mongol violence was not simply the product of barbarism or impulse, but a strategic adaptation to their comparative advantages and institutional constraints.

A. Proofs

Proof of Proposition 1. If D_j cooperates, D_i ’s best response is to cooperate if

$$(1 - \tau_i)s_i \geq p(m_i, m_f + m_j) - c_i,$$

$$\tau_i \leq 1 - \frac{p(m_i, m_f + m_j) - c_i}{s_i} \equiv \tau_i^C,$$

and to resist otherwise. Similarly, if D_j resists, D_i ’s best response is to cooperate if

$$p(m_f + m_i, m_j)(1 - \tau_i) - c_i \geq p(m_1 + m_2, m_f)s_i - c_i,$$

$$\tau_i \leq 1 - \frac{p(m_1 + m_2, m_f)s_i}{p(m_f + m_i, m_j)} \equiv \tau_i^R,$$

and to resist otherwise.

If $\tau_i \leq \tau_i^C \forall i = 1, 2$, both factions cooperate. If $\tau_1 \leq \tau_1^R$ and $\tau_2 > \max\{\tau_2^C, \tau_2^R\}$, D_1 cooperates and D_2 resists. If $\tau_2 > \max\{\tau_1^C, \tau_1^R\}$ and $\tau_1 \leq \tau_1^R$, D_1 resists and D_2 cooperates. If $\tau_i > \max\{\tau_i^C, \tau_i^R\}$, both resist.

If both cooperate, F receives $\tau_1 s + \tau_2(1-s)$, increasing in both τ_1 and τ_2 . It is best for F to propose (τ_1^C, τ_2^C) which will be accepted, and

$$\begin{aligned} EU_F(\tau_1^C, \tau_2^C) &= \sum_{i=1,2} \left(1 - \frac{p(m_i, m_f + m_j) - c_i}{s_i} \right) s_i \\ &= 1 - p(m_1, m_f + m_2) - p(m_2, m_f + m_1) + c_1 + c_2 \\ &= p(m_f, m_1 + m_2) + c_1 + c_2. \end{aligned}$$

If D_1 cooperates and D_2 resists, F receives $p(m_f + m_1, m_2)\tau_1 - c_f$, which increases in τ_1 . The best F can propose to induce D_1 's cooperation is τ_1^R . And F can propose any $\tau_2 > \max\{\tau_2^C, \tau_2^R\}$ that guarantees resistance by D_2 . This will give F

$$\begin{aligned} EU_F(\tau_1) &= p(m_f + m_1, m_2) \left[1 - \frac{p(m_1 + m_2, m_f)s}{p(m_f + m_1, m_2)} \right] - c_f \\ &= p(m_f + m_1, m_2) - p(m_1 + m_2, m_f)s - c_f, \end{aligned}$$

which is larger than $p(m_f, m_1 + m_2) + c_1 + c_2$ if

$$\begin{aligned} p(m_f + m_1, m_2) - p(m_1 + m_2, m_f)s - c_f &> p(m_f, m_1 + m_2) + c_1 + c_2 \\ s &< \frac{p(m_f + m_1, m_2) - p(m_f, m_1 + m_2) - (c_f + c_1 + c_2)}{p(m_1 + m_2, m_f)} \\ s &< p(m_1, m_2) - \frac{c_f + c_1 + c_2}{p(m_1 + m_2, m_f)} = \underline{s}. \end{aligned}$$

Thus if $s < p(m_1, m_2) - \frac{c_f + c_1 + c_2}{p(m_1 + m_2, m_f)}$, F offers $\tau_1 = \tau_1^R$ and any $\tau_2 > \max\{\tau_2^R, \tau_2^C\}$; D_1 cooperates and D_2 resists, establishing the first case. The same process for the case of D_1 resisting and D_2 cooperating yields that if $s > p(m_1, m_2) + \frac{c_f + c_1 + c_2}{p(m_1 + m_2, m_f)} = \bar{s}$, F offers $\tau_2 = \tau_2^R$ and any $\tau_1 > \max\{\tau_1^R, \tau_1^C\}$; D_2 cooperates and D_1 resists, establishing the second case. If $p(m_1, m_2) - \frac{c_f + c_1 + c_2}{p(m_1 + m_2, m_f)} < s < p(m_1, m_2) + \frac{c_f + c_1 + c_2}{p(m_1 + m_2, m_f)}$, F 's maximum expected utility from inducing both factions to cooperate is larger than F 's maximum expected utility from inducing one faction to cooperate and the other to resist. Thus, F offers (τ_1^C, τ_2^C) , and both D_1 and D_2 cooperate, establishing the third case.

If there is mutual resistance, it must be the case that the minimum offer for the peace premium for both factions is lower than 0. This happens if

$$p(m_i, m_f + m_j) - c_i > s_i \quad \forall i \in \{1, 2\}.$$

Solving this for both factions gives the following system of inequalities:

$$s < f(m_1, m_2 + m_f) - c_1, \quad s > f(m_f + m_1, m_2) + c_2.$$

Since the latter condition is strictly larger than the former, the two equations can never be true simultaneously. Therefore, mutual resistance is impossible; the Foreign always has a viable offer.

The cutpoints on both sides exist if and only if $p(m_1, m_2) - \frac{c_1+c_2+c_f}{p(m_1+m_2, m_f)} > 0$ and $p(m_1, m_2) - \frac{c_1+c_2+c_f}{p(m_1+m_2, m_f)} < 1$. This gives us the result that

$$1 - \frac{c_1 + c_2 + c_f}{p(m_1 + m_2, m_f)} < p(m_1, m_2) < \frac{c_1 + c_2 + c_f}{p(m_1 + m_2, m_f)}. \quad \square$$

Proof of Proposition 2.

$$EU_F(\tau_1^C, \tau_2^C) = \sum_{i=1,2} \left(1 - \frac{p(m_i, m_f + m_j) - c_i}{s_i} \right) s_i = p(m_f, m_1 + m_2) + c_1 + c_2 > p(m_f, m_1 + m_2)(1 - \kappa_D) - c_f. \quad \square$$

Proof of Lemma 1. D_i 's best response is to cooperate if

$$p(m_f + m_i, m_j)(1 - \tau_i) \geq p(m_1 + m_2, m_f)s_i - c_i,$$

$$\tau_i \leq 1 - \frac{p(m_1 + m_2, m_f)s_i}{p(m_f + m_i, m_j)},$$

and to resist otherwise. The parameter space under which one faction resists is determined. If D_j resists, the rate at which the Foreign can possibly commit to offer D_i once D_j is out of the picture for this faction is $p(m_i, m_f)c_i$. Therefore, it is pointless to offer more than that. Thus,

$$\tau_i^R = \min \left\{ p(m_f, m_i) + c_i, 1 - \frac{p(m_1 + m_2, m_f)s_i}{p(m_f + m_i, m_j)} \right\}.$$

Then we could observe that the war equivalent payoff is higher if and only if

$$1 - \frac{p(m_1 + m_2, m_f)s_i}{p(m_f + m_i, m_j)} > p(m_f, m_i) + c_i \iff s_i < p(m_1, m_2) + c_i \cdot \frac{m_f + m_1}{m_1 + m_2}.$$

Since $p(m_1, m_2) + c_i \cdot \frac{m_f + m_1}{m_1 + m_2} > p(m_1, m_2) - \frac{c_f + c_1 + c_2}{p(m_1 + m_2, m_f)} = \bar{s}$, the offer that successfully collaborates with the other faction is always more stringent than its war payoff. The commitment issue arising from intra-coalition instability will not hurt the enforceability of the deal. $\square$

Proof of Proposition 3. The expected payoff for the unilateral invasion of the Foreign becomes

$$EU_F(I) = p^*(\lambda m_f, m_1 + m_2)(1 - \kappa_D) - c_f = \frac{\lambda m_f(1 - \kappa_D)}{\lambda m_f + m_1 + m_2} - c_f.$$

Foreign chooses to co-opt with two factions, where there exists $s_1^* = p(m_1, m_2) - \frac{c_f + c_1 + c_2}{p(m_1 + m_2, m_f)}$, $s_2^* = p(m_1, m_2) + \frac{c_f + c_1 + c_2}{p(m_1 + m_2, m_f)}$ such that $s \in (s_1^*, s_2^*)$; chooses to co-opt with D_1 when $s \in (0, s_1^*)$ and chooses to co-opt with D_2 when $s \in (s_2^*, 1)$.

Now if $s \in (0, s_1^*)$, D_1 cooperates and D_2 resists, F receives $1 - p(m_1 + m_2, m_f)s - c_f$. Foreign chooses to unilaterally invade if and only if

$$\kappa_D < 1 - \frac{\left(\frac{m_f + m_1}{m_f + m_1 + m_2} - \frac{m_1 + m_2}{m_1 + m_2 + m_f} s \right) (\lambda m_f + m_1 + m_2)}{\lambda m_f} = \kappa_D^\dagger, \quad \text{or} \quad (3)$$

$$\lambda > \frac{\left(\frac{m_f + m_1}{m_1 + m_2 + m_f} - \frac{m_1 + m_2}{m_1 + m_2 + m_f} s \right) (m_1 + m_2)}{m_f \left(1 - \kappa_D - \left(\frac{m_f + m_1}{m_1 + m_2 + m_f} - \frac{m_1 + m_2}{m_1 + m_2 + m_f} s \right) \right)} = \lambda^\dagger. \quad (4)$$

Similarly, when $s \in (s_2^*, 1)$, D_2 cooperates and D_1 resists, F receives $1 - p(m_1 + m_2, m_f)(1 - s) - c_f$. Foreign chooses unilateral invasion if and only if

$$\kappa_D < 1 - \frac{\left(\frac{m_f + m_2}{m_f + m_2 + m_1} - \frac{m_1 + m_2}{m_1 + m_2 + m_f}(1 - s)\right)(\lambda m_f + m_1 + m_2)}{\lambda m_f} = \kappa_D^\dagger, \quad \text{or} \quad (5)$$

$$\lambda > \frac{\left(\frac{m_f + m_2}{m_f + m_2 + m_1} - \frac{m_1 + m_2}{m_1 + m_2 + m_f}(1 - s)\right)(m_1 + m_2)}{m_f \left(1 - \kappa_D - \left(\frac{m_f + m_2}{m_f + m_2 + m_1} - \frac{m_1 + m_2}{m_1 + m_2 + m_f}(1 - s)\right)\right)} = \lambda^\dagger. \quad (6)$$

The substantive story for both of these scenarios is similar. Foreign compares the cost of direct administration to the share of spoil that the cooperating faction occupies (s_i). If the cost of administrative cost is significantly lower than s_i , then it might be possible that unilateral invasion is rational for Foreign when first-strike advantage is high.

If the domestic forces are sufficiently balanced such that $s \in (s_1^*, s_2^*)$, Foreign now chooses to cooperate with both factions, and it would receive $1 - p(m_1, m_f + m_2) - p(m_2, m_f + m_1) + c_1 + c_2$. Foreign chooses to unilaterally invade if and only if

$$\kappa_D < 1 - \frac{\left(\frac{m_f}{m_2 + m_f + m_1} + c_1 + c_2\right)(\lambda m_f + m_1 + m_2)}{\lambda m_f} = \tilde{\kappa}_D, \quad \text{or} \quad (7)$$

$$\lambda > \frac{\left(\frac{m_f}{m_2 + m_f + m_1} + c_1 + c_2\right)(m_1 + m_2)}{m_f \left(1 - \kappa_D - \left(1 - \frac{m_1}{m_1 + m_f + m_2} - \frac{m_2}{m_2 + m_f + m_1} + c_1 + c_2\right)\right)} = \tilde{\lambda}. \quad \square$$

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